

The Hidden Bias in AI

How Heart Scans Can Unintentionally Reveal Race

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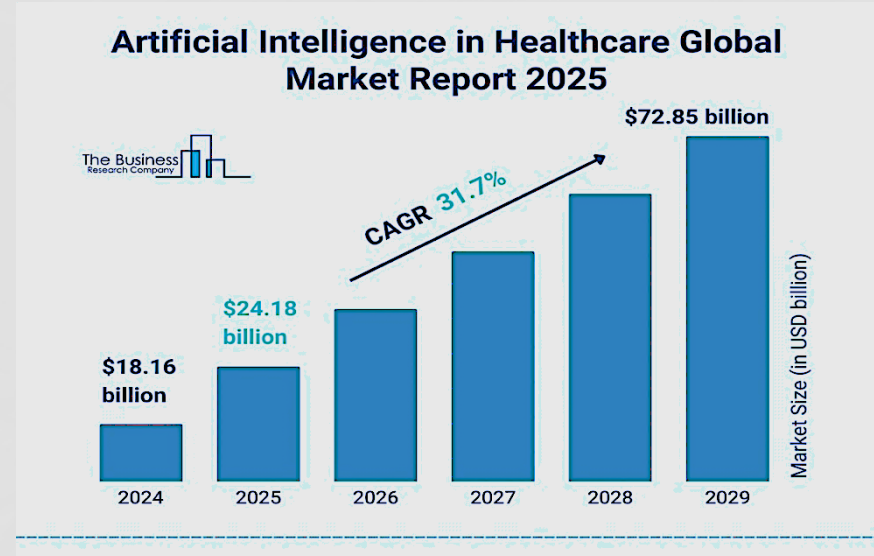
June 8, 2025

Based on: Lee, T. (2025, May 12). AI can guess racial categories from heart scans – what it means and why it matters. The Conversation.



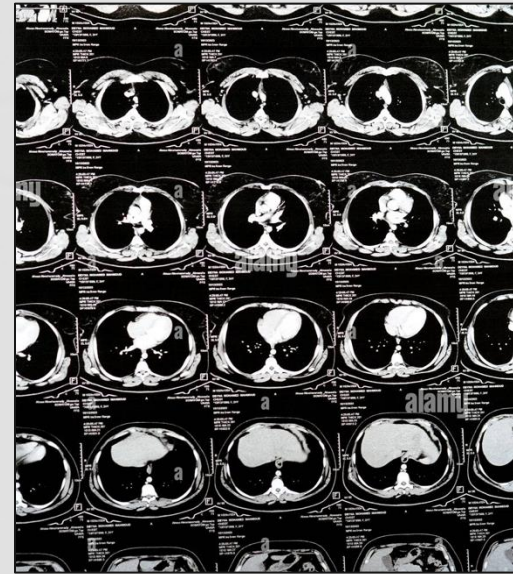
AI in Healthcare:

- AI is transforming healthcare (analysis, prediction, personalization).
- **Key Issue:** AI must be free from social biases like race for fair and effective healthcare.



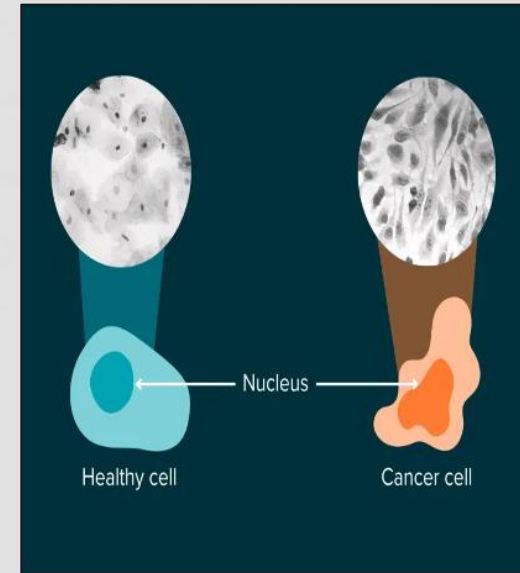
AI Predicts Race from Heart Scans

- Study: AI predicted Black/white identity from heart scans *ALONE*, with 96% accuracy.
- No explicit training on race was provided to the AI.
- AI isn't a neutral observer; it learns hidden patterns linked to social categories.



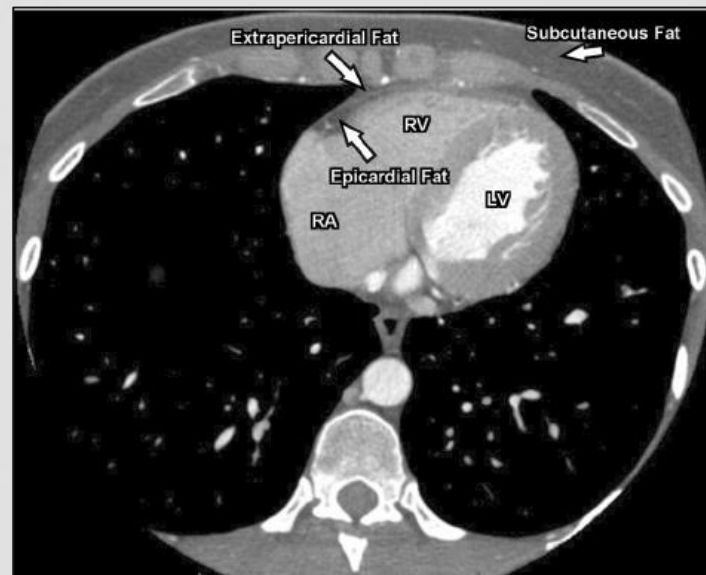
The Social Construct of Race and AI

- AI models often excel at classification based on genuine biological differences.
- Race is not a fundamental biological difference in the same way diseases or physiological conditions are.
- Race is a social construct.



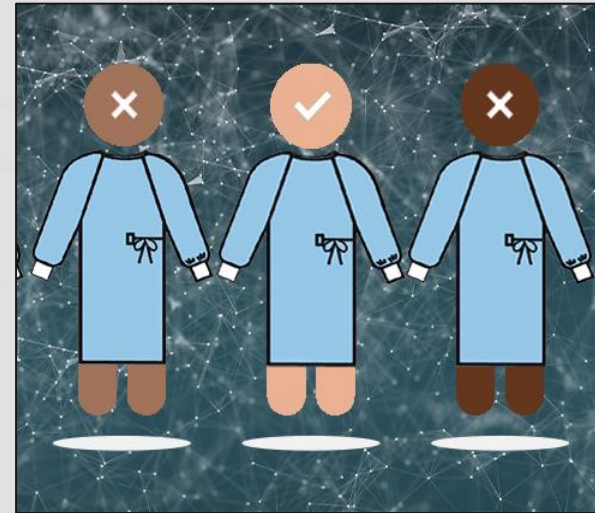
AI's Deceptive Shortcuts in Heart Scans

- AI was not analyzing the heart itself for biological racial markers.
- AI used subcutaneous fat instead!
- "Shortcut learning": Easy but incorrect associations with race.
- What if AI is using Shortcut learning in other models.



The Danger of using AI in Healthcare

- AI is a black box:
It may use shortcut learning and analyze irrelevant data.
- AI is not a neutral observer:
Potential for unequal treatment and worsened health disparities.



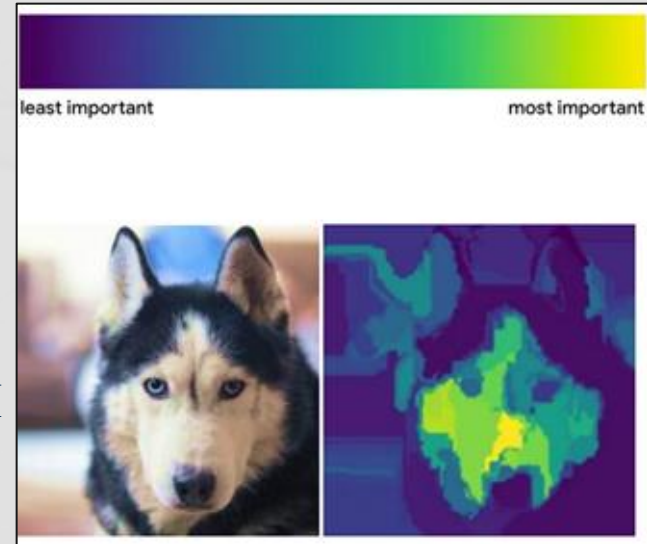
Solutions for Fair and Transparent AI in Healthcare

Diversify training data:

Ensure equal representation of all populations to avoid bias from traits like subcutaneous fat that vary across groups.

Increase transparency:

Use Explainable AI (e.g., heat maps) to reveal AI decisions and detect bias.



Conclusion

- AI is revolutionizing healthcare but is not free from bias.
- This raises serious concerns about equity, accuracy, and trust in medical AI.
- Shortcut learning makes AI systems prone to unintended and harmful assumptions.
- Ensuring fair, transparent, and explainable AI is a must.



Thank You for Listening

